Response Letter with Annotations
Jiakun Liang

Annotations

1. **Purposes**
   a. Motivation (one-sentence summary): The study examines the hierarchical structures of communication networks on Twitter and their roles in increasing the reach of protest messages.
   b. Specific Rationale (several bullet points):
      - The role of social media in political mobilization remains controversial, with comparison between daring minority of highly committed protesters and the followers with relatively little engagement in meaningful participation.
      - Resource mobilization and critical mass theories emphasize that the success of collective action depends on the ability to disseminate information and scalable participation, yet existing research has largely focused on the number of participants mobilized rather than communication dynamics.
      - Existing research on how social networks mediate collective actions has derived from stimulated or mathematical models, which did not capture the complex core-periphery structure seen in many empirically observed networks.
   c. Research Questions and/or Hypotheses
      - Identify IV, DV, mediator, moderator
        IV: k-score level in communication networks
        DV: activity and reach
        Moderator: protest vs. non-protest messages
      - Research questions:
        RQ1: What is the hierarchical structure of protest-related communication in Twitter networks?
        RQ2: What are the roles of core and peripheral in the diffusion of protest messages?
        RQ3: How do communication networks of protest messages differ from non-protest messages?

2. **Method**
   a. Research Design:
      Social network analysis using k-core decomposition to identify the core-periphery structure diffusion networks of protest messages.
   b. Sample (representative sample?)
      Five datasets in Twitter (3 protest networks and 2 non-protest networks, e.g., Gezi dataset: 30,019,710 tweets from 2,908,926 users)
   c. Measures
      - **Activity:** Total number of protest-related tweets posted by participants (including original tweets and retweets).
      - **Reach:** The proportion of total followers exposed to protest-related information.

- **Node:** one individual unit
- **Degree:** number of retweets made or received by one node
- **Network structure:** Measured using k-core decomposition, which identifies groups of users by progressively trimming the least connected nodes in a network.

d. Analysis:

- Construct retweet networks.
- Use k-core decomposition to classify users into core and peripheral layers.
- Measure activity and reach contributed by each k-core.
- Simulate the removal of peripheral users to evaluate their impact on information diffusion.
- Compare results with a random benchmark where k-core labels are randomly assigned which can be interpreted as a line of perfect equality, that is, what would happen if all k-cores contribute the same amount to overall activity and reach.

3. Results and Conclusions

a. Answers to RQs and Hypotheses

- **RQ1:** A small portion of core users produces most information, while a large portion of peripheral users mainly disseminates messages.
- **RQ2:** Removing peripheral users leads to a dramatic reduction in overall reach.
- **RQ3:** The core-periphery pattern is especially strong in protest networks but much weaker in non-protest networks.

b. Include Key Figures/Tables

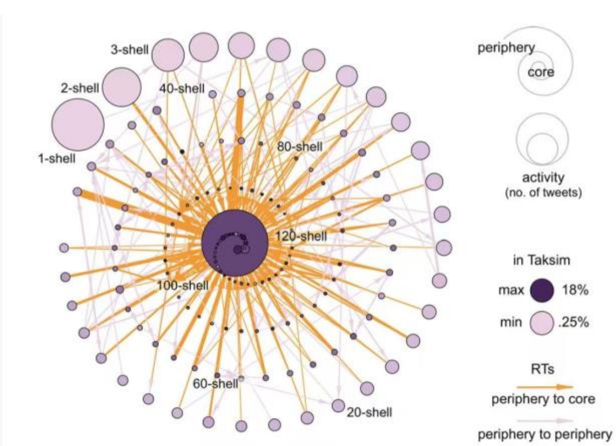
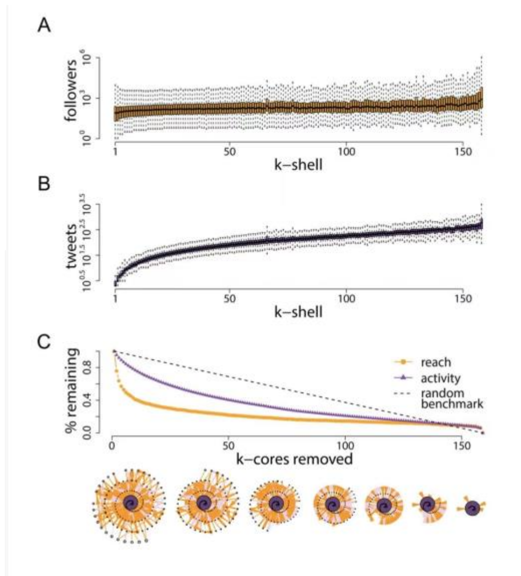


Fig 3. *K*-core decomposition of the network of retweets that emerged during the 2013 Taksim Gezi Park protests in Turkey (see S1 Text). Participants have been grouped in their corresponding *k*-shells, here represented by nodes. Lower *k*-shells contain participants at the periphery of the network; higher *k*-shells contain core participants. Node size is proportional to aggregated activity, measured as total number of protest messages (not just retweets). Arcs indicate retweeting activity, and their width is proportional to normalized strength (arcs with lower strength have been filtered to improve the visualization of the network). The darkness of nodes is proportional to the percentage of participants who reported being in the Taksim Gezi Park (the geographical epicenter of the protests), as indicated by the geographic information attached to their tweets. Most of these participants are at the core of the network where most RTs are also sourced from, thus allowing information to flow from the core to the periphery.



PLOS ONE | DOI:10.1371/journal.pone.0143611 November 30, 2015 8 / 15

PLOS ONE The Critical Periphery in Social Protests

Fig 4. Audience size and activity levels across *k*-cores. Panel A shows the distribution in number of followers (or reach) across *k*-shells. Panel B plots the distributions in number of tweets sent (or activity) across k-shells. Panel C shows the effects on overall reach and activity of removing *k*-cores progressively, starting from the lowest or most peripheral as illustrated by the networks below the horizontal axis. Removing the first five *k*-cores results in a drop of slightly more than 50% in total reach capacity, suggesting that the sphere of influence of core participants is much reduced without peripheral contributors. The random benchmark is based on 10,000 permutations of the data where assignment to *k*-cores is randomly re-shuffled; this benchmark can be interpreted as a line of perfect equality, i.e. a scenario in which all *k*-cores contribute the same amount to overall activity and reach.

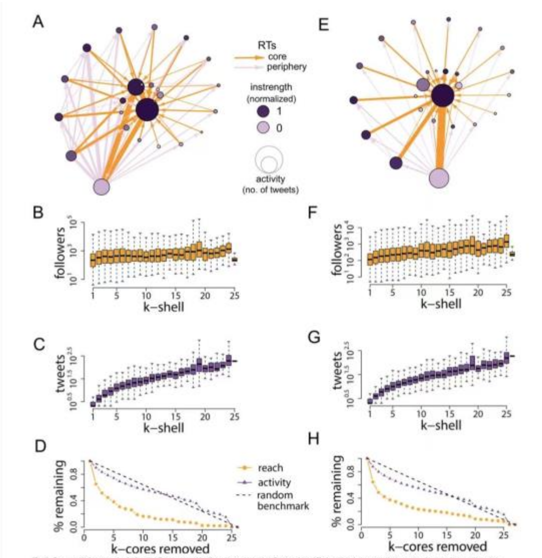


Fig 5. Core-periphery analysis for the Occupy and Indignados networks. Panels A and E visualize the connections across k-cores (arcs with lower strength have been filtered to improve visualization). Unlike the Turkish case, these protests did not have a clear epicenter, so nodes are colored in proportion to the number of retweets received (i.e. normalized instrength). Core-periphery dynamics are, however, similar to the Turkish case: most of the

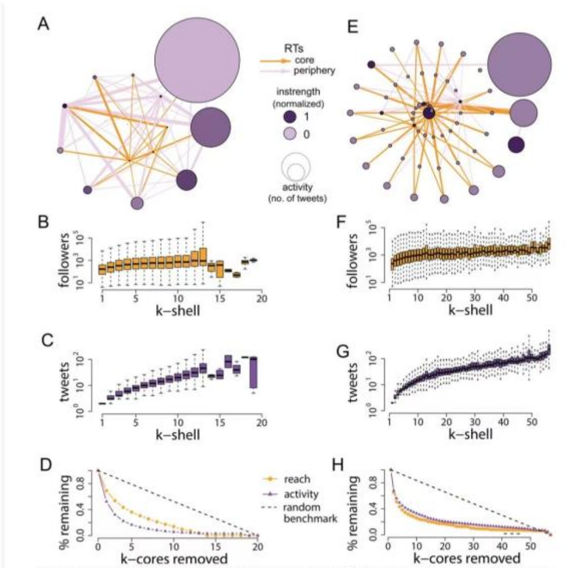


Fig 6. Audience size and activity levels across k-cores for the Oscars network (panels A-D) and the minimum wage network (panels E-H). Arcs with lower strength have been filtered to improve network visualization.

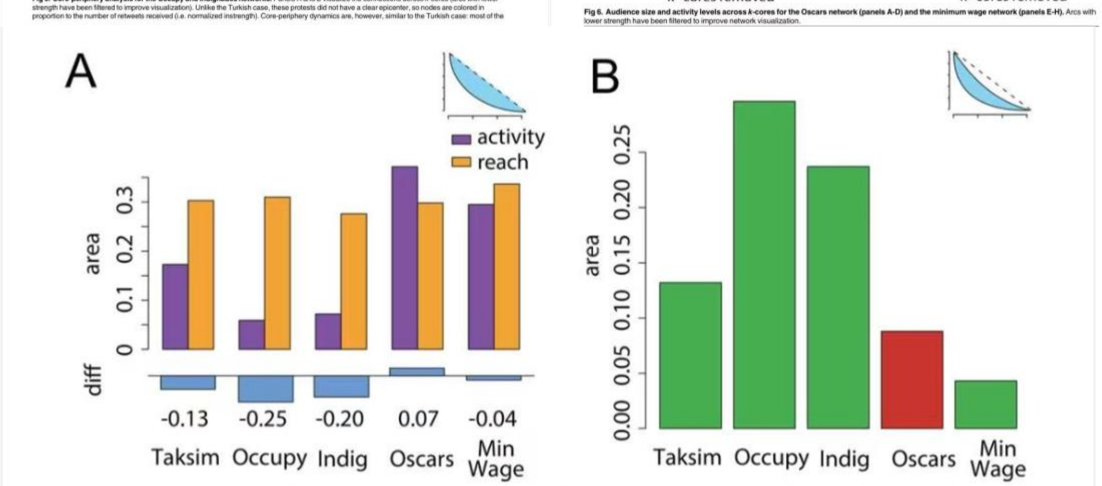


Fig 7. Area above reach and activity curves, bounded by random benchmark (panel A) and area between reach and activity curves, which serves as an index of core-periphery dynamics (panel B). The diagrams on the upper-right corner indicate the areas that were computed in each panel. The Oscars dataset is the only one in which removing the lower cores has a greater impact on activity than reach–hence its different color in the barplot of panel B.

c. Authors' Conclusions

Twitter protest communication networks show a strong core-periphery structure, and peripheral users contribute little activity but collectively play a crucial role in expanding the audience reach of protest messages.

4. Discussion

a. Implications: The study challenges the idea that only core users drive information diffusion. Instead, peripheral users play an important role in expanding the reach of messages.

b. Limitations:

❿ It remains unclear whether the specific characteristics of the events (e.g., geographical location, nature of protests, disruptive force) made them uniquely effective at activating a large periphery.

❿ Core-periphery dynamics are the result of exogenous factors shaping the process, which limits the ability to generalize findings across different contexts.

- ❿ The study is still far from understanding how to best characterize these connections and how network structure relates to cascading phenomena in collective action settings.

c. Future Directions

- ❿ Future work should consider alternative data sets to test whether the findings hold across different contexts.
- ❿ Future research should further investigate how the distributions of influential and susceptible people and their networks shape collective actions.

--- Page Break Here ---
Response Letter (~600 words)

- Your assessment of the study
- Weaknesses
- Other notes

Barberá et al. (2015) examine how information spreads through Twitter networks during political protests by analyzing large-scale communication data from multiple protest events. Using retweet networks and k-core decomposition, the authors identify a clear core-periphery structure in which a small and highly interconnected group of core participants generates much of the protest messages, while a large group of peripheral users with less engagement plays an important role in amplifying its reach. By comparing the activity and reach after removing peripheral users with a random simulated benchmark, the study demonstrates that peripheral participants, despite their low individual influence, collectively contribute to a substantial audience reach. Furthermore, comparisons with non-protest networks reveal that the roles of core and periphery is more pronounced in protest contexts.

A key strength of the methodology is the use of large-scale communication data on Twitter to identify the core-periphery pattern of diffusion of protest messages in social networks. Specifically, by analyzing millions of tweets and constructing retweet networks, the study captures how messages are disseminated during protests in real world that cannot be achieved by survey or experimental data, enhancing the ecological validity and robustness of the results. Another important methodological innovation is the use of k-core decomposition to distinguish between core and peripheral users. Different traditional measures that focus on individual level of social networks, such as degree, network size, and density, k-core captures the more structural and hierarchical networks, identifying the relative position of users within the interconnected groups. I think this is particularly useful for studying collective action, where influence depends not only on individual reach but also the interconnectivity of individuals within the networks.

The study further strengthens its methodological rigor through the use of simulation-based analysis. By simulating how removing them from the network would affect activity and reach and comparing the resulting against a random benchmark, which is based on a random assignment of k-core values over times to construct a line of perfect equality, the methodology establishes a baseline for assessing structural contribution of the periphery, as the removal of any hierarchy will systematically weaken the activity and reach anyway. This design is particularly effective in demonstrating that the observed impact of peripheral users is not merely a byproduct of network size or random variation, but rather a consequence of the network's core-periphery organization.

The study also has its limitations. First, as noted by the authors, the core-periphery structure in communication networks is the result of exogenous factors shaping the process, which limits the ability to generalize the findings across different contexts. For instance, in China, state censorship and platform regulations may constrain the information flow and reshape network structures, where even retweeting can bring about potential risks, such as account block, which may discourage participation and bias the observed diffusion patterns. Second, the study focuses on the roles the core and peripheral individuals play in disseminating protest-related information, which is far from understanding the how network structure relates to cascading phenomena in。For instance, we only know that peripheral users collectively contribute to the amplification of information diffusion, but not the specific mechanisms through which why they initiate the cascading process and diffuse the information across the social networks.

Even though it is innovative to use the k-core decomposition to identify core and peripheral participants, I am confused about whether this method truly reflects the roles of participants in the diffusion process, rather than their structural positions. For instance, if I understand correctly, according to the method, a journalist with little interconnectivity within his networks but still be retweeted by a substantial number of users may be classified as peripheral, whereas a highly interconnected group (e.g., friends) may retweet each other but with limited actual influence on information diffusion may be classified core.

In conclusion, Barberá et al. (2015) provide an insightful theoretical contribution by demonstrating the importance of peripheral users in information diffusion within protest networks through an innovative social network analysis approach. The use of k-core decomposition and simulation offers meaningful insights into the structural dynamics of online diffusion of protest-related messages. However, the study remains limited in its ability to explain the underlying mechanisms that drive cascading processes and generalizability to other contexts. Future research should integrate more behavioral and contextual factors to better understand how information spreads in complex social networks.